Mitra · sovereign English full-duplex voice · models to load, and what Gemini costs

27 Oct 2026 · Sakala Mitra, Government of Karnataka · For the AI ops team · Scope: English only, every model free to download, licences that allow a government to run them in production, nothing leaves our hardware. Gemini Live stays the measured reference until the core matches it; it is never a training source. Hardware: the two H200s in the one office server.

Part 1 · Every model the English sovereign architecture needs

The architecture is the one in the 90-day plan with one change for English: the text brain becomes **gpt-oss-120b** instead of Sarvam, because it is Apache 2.0, stronger in English, and fits beside the duplex core on one card. One native duplex model hears and speaks at the same time; the existing Mitra router, SQL and tools decide; results are pushed back into the core and spoken in its own voice. No separate speech-to-text, text model and text-to-speech in the voice path.

1.1 · The serving stack: what runs on the two cards at all times

Role	Model, Hugging Face id	Size on disk	Licence and condition	Why this one
Duplex core, primary	`nvidia/personaplex-7b-v1`	7B, bf16, about 16 GB; gated, accept the terms on download	NVIDIA Open Model License; the card says "ready for commercial use". Base Moshi weights CC-BY-4.0, so Kyutai is credited in the product notice	Moshi with persona and voice prompts already trained in. Full-Duplex-Bench: interruption takeover 0.95 at 0.24 s, smooth turn-taking 0.908 at 0.17 s. Gemini Live measures 1.2 s.
Duplex core, base and fallback	`kyutai/moshiko-pytorch-bf16` `kyutai/moshika-pytorch-bf16`	7B plus depth transformer, about 16 GB each	CC-BY-4.0 weights; Python code MIT, Rust server Apache 2.0	The reference architecture. Needed anyway: PersonaPlex is built on Moshiko, and moshi-finetune trains on it.
Audio codec	`kyutai/mimi`	96M, under 0.5 GB	CC-BY-4.0	12.5 Hz, 24 kHz, 1.1 kbps. Bundled with the core. The user transcript branch and the factory use it too.
Live user transcript	`kyutai/stt-1b-en_fr`	1B, about 2 GB	CC-BY-4.0	Streaming, 0.5 s delay. Moshi's text stream covers only Mitra's own words, so the citizen's words for the dispatcher come from this.
Text brain	`openai/gpt-oss-120b`	117B total, 5.1B active, MXFP4, about 65 GB, fits one 80 GB card	Apache 2.0	Picks the intent as guided JSON, words difficult replies, reasons. Runs on vLLM.
Text brain, fast fallback	`Qwen/Qwen3-30B-A3B`	30B total, 3B active, FP8 about 31 GB	Apache 2.0	If the 120B is busy or memory is tight on the training nights.
Voice activity	`snakers4/silero-vad`	2 MB	MIT	The relay's echo gate and silence watchdog.
End-of-turn safety net	`pipecat-ai/smart-turn-v3`	about 30 MB, ONNX	BSD-2-Clause	Semantic end-of-turn in the safety controller for the harness and the overflow path; English covered.
Noise suppression	DeepFilterNet3, `Rikorose/DeepFilterNet`	about 10 MB	MIT or Apache 2.0, our choice	Cleans the 16 kHz mic stream before the core when the room is noisy; runs on CPU.
Wake word	openWakeWord, `dscripka/openWakeWord`, with a "Hey Mitra" model we train	a few MB, CPU	Code Apache 2.0. The pre-trained example models are CC-BY-NC-SA, so we do not use those; a model we train ourselves is ours	Replaces Gemini's transcription of every room utterance in Standby. Trained from synthetic speech.
Wake-word training voice	Piper, `rhasspy/piper`	about 60 MB per voice	MIT	Generates the thousands of "Hey Mitra" clips openWakeWord needs.

1.2 · The factory and evaluation stack: offline, on the same cards at night

Role	Model	Size	Licence	Why
Fine-tuning	`kyutai-labs/moshi-finetune`	code	Apache 2.0	LoRA or full tuning of Moshi and PersonaPlex. Their figure: one H100, 39.6 GB peak. Data is stereo WAV, Mitra left, user right, with JSON transcripts it can make itself.
Synthetic dialogue voices	`kyutai/tts-1.6b-en_fr` with `kyutai/tts-voices`	1.8B, about 4 GB	CC-BY-4.0; cloning limited to the published voice embeddings	Streaming, batch throughput 75× real time. Voices the scripted Sakala dialogues in two voices. Same Mimi codec as the core.
Second synthesis voice, for variety	`ResembleAI/chatterbox`	500M, about 2 GB	MIT	Different voice family for the citizen side; emotion control.
Expressive multi-speaker synthesis	`microsoft/VibeVoice-1.5B`	3B, about 6 GB	MIT, but the card says research use only	Up to four speakers in one long recording; use only for evaluation material, not production.
Conversational TTS, research	`sesame/csm-1b`, `canopylabs/orpheus-3b-0.1-ft`	2B and 4B	Apache 2.0, both gated	Alternatives for the citizen voice in synthetic data; Orpheus streams at about 200 ms.
Scoring ASR, batch	`nvidia/canary-qwen-2.5b` and `nvidia/parakeet-tdt-0.6b-v3`	5 GB and 2.5 GB	CC-BY-4.0	Filters synthetic dialogues by word error and scores the core's output. Both top the open ASR leaderboard at about 6% WER.
Scoring ASR, streaming reference	`nvidia/nemotron-speech-streaming-en-0.6b`	2.5 GB	NVIDIA Open Model License; verify the card	What Full-Duplex-Bench style timing measurements transcribe with.

Independent ASR	`openai/whisper-large-v3-turbo`	1.6 GB	MIT	Second opinion in the filters so one recogniser's bias does not shape the data.
Word timestamps	`Qwen/Qwen3-ForcedAligner-0.6B`	1.2 GB	Apache 2.0	Aligns transcripts to audio at 43 ms average error for the inner-monologue training targets.
Speaker separation	`pyannote/speaker-diarization-community-1`	50 MB, gated	CC-BY-4.0	Splits any mixed recording into the two channels the trainer needs. Apache alternative: the `diarize` package.
Benchmark	Full-Duplex-Bench, `DanielLin94144/Full-Duplex-Bench`	code	licence file in the repo, read it	v1.0 pause handling, backchannel, turn-taking, interruption; v1.5 overlaps; v3.0 tool use under disfluent speech. The public yardstick PersonaPlex and Gemini are scored on.
Dialogue scripts	gpt-oss-120b, already loaded		Apache 2.0	Writes the Sakala conversations with interruptions, repairs and backchannels for the factory.
Guard, optional	`ibm-granite/granite-guardian-3.3-8b`	16 GB	Apache 2.0	A text safety and injection classifier behind our regex guard. Not needed for the sprint.

1.3 · The data: all free and licensed for a government product

Corpus	What	Hours	Terms
Open Yap 1K, The Agentic Data Company	Natural two-speaker English conversations, each speaker on its own track, 48 kHz, 239 consented and paid speakers	1,000 h in 1,602 conversations	Free under its Data Use Agreement: commercial use, training and deployment allowed; no redistribution, no identifying speakers, no voice clones. Access is requested on their site and granted within hours. The 8.9 h sample is CC-BY-4.0 on Hugging Face.
DailyTalk, `kyutai/DailyTalkContiguous`	Scripted two-speaker dialogue in moshi-finetune's format	14 GB	CC-BY-SA 4.0
AMI Meeting Corpus	Multi-speaker English meetings, close-talk mics	100 h	CC-BY-4.0
Our own synthetic Sakala dialogues	Scripts by gpt-oss-120b, voiced by Kyutai TTS and Chatterbox, filtered by Canary and Whisper	2,000 to 5,000 h, made on our cards	Ours
MUSAN, RIRS_NOISES	Noise, music, room impulse responses for augmentation		CC-BY-4.0, Apache 2.0

> **Two models researched and not recommended.** NVIDIA NemotronLabs VoiceChat 11B has the native tool-call stream we want, but its card puts it under the OpenMDW 1.1 licence and describes it as targeted at researchers and developers; secondary sources read it as research-only. Until legal reads that licence, it stays out. Sarvam-105B is Apache 2.0 and the right brain for Kannada later, but its FP8 build needs both cards, which this English sprint cannot give it.

1.4 · Placement on the two H200s

GPU 1 · the duplex worker

PersonaPlex core, Mimi, Kyutai STT 1B, Silero, Smart Turn: about 22 GB of models and 0.6 GB per live session, so 100 to 150 sessions in memory.

GPU 2 · the brain, a second worker, the factory by night

gpt-oss-120b at 65 GB plus 20 GB of cache, a second duplex worker at 16 GB, and the factory models by night; moshi-finetune's 40 GB fits by pausing the second worker.

Total downloads about 230 GB. Gated repositories (PersonaPlex, CSM, Orpheus, pyannote) need a Hugging Face account that has accepted each model's terms; the Open Yap 1K request is made on the publisher's site and is granted within hours.

1.5 · Download order for AI ops, day one

1. `nvidia/personaplex-7b-v1` (gated), `kyutai/moshiko-pytorch-bf16`, `kyutai/moshika-pytorch-bf16`, `kyutai/mimi`
2. `kyutai/stt-1b-en_fr`, `kyutai/tts-1.6b-en_fr`, `kyutai/tts-voices`
3. `openai/gpt-oss-120b`, `Qwen/Qwen3-30B-A3B` (FP8 build), vLLM
4. `nvidia/canary-qwen-2.5b`, `nvidia/parakeet-tdt-0.6b-v3`, `openai/whisper-large-v3-turbo`, `Qwen/Qwen3-ForcedAligner-0.6B`, `pyannote/speaker-diarization-community-1` (gated)
5. `snakers4/silero-vad`, `pipecat-ai/smart-turn-v3`, DeepFilterNet3, openWakeWord code with Piper, `ResembleAI/chatterbox`
6. Code: `kyutai-labs/moshi` (Rust server), `kyutai-labs/moshi-finetune`, `DanielLin94144/Full-Duplex-Bench`
7. Data: the Open Yap 1K access request, DailyTalk, AMI, MUSAN, RIRS_NOISES

Legal reads the NVIDIA Open Model License before day 1; the answer is clean either way because Moshiko is CC-BY-4.0. Every CC-BY model is credited in the product notice. Nothing in this list phones home; every model is a file we hold on the office server.

Part 2 · What Gemini costs now and at scale

Everything below is the metered cost of running Mitra on Gemini. It is what the sovereign stack in Part 1 replaces for English. One rupee figure throughout: **$1 = ₹95**, the observed rate in August and September 2026.

2.1 · The rate card, paid tier, from Google's pricing page on 27 Oct 2026

Model	Text in	Text out	Audio in	Audio out
Gemini 3.8 Live, the voice	$0.75 /M	$4.50 /M	$3.00 /M, $0.005 per minute	$12.00 /M, $0.018 per minute
Gemini 3.8 Flash and 3.6 Flash	$0.75 /M, rising to $1.50 on 1 Jan 2027	$3.75 /M, rising to $7.50		
Gemini 3.1 Flash-Lite	$0.25 /M	$1.50 /M	$0.50 /M	
Gemini 3.8 Flash TTS	$0.50 /M, rising to $1.00			$9.00 /M, rising to $18.00

Audio is billed at 25 tokens a second. The Live session is billed for the whole time it is open, whether or not anyone is speaking.

2.2 · What one Mitra conversation costs

From our own logs, a live conversation averages 8.5 user turns, nearly every one with a function call. The unit is a 5-minute conversation with the citizen and Mitra each speaking about a third of the time.

Item	Tokens or minutes	Cost
Live audio in: the whole open session is billed	5 min	$0.025
Live audio out: Mitra speaking	1.75 min	$0.032
Live text: instruction, function declarations, results; about 8k tokens a turn if Google re-bills the context each turn, as developers report	68k tokens	$0.051, or $0.008 if the context is not re-billed
POC calls behind the filing: understanding on 3.6 Flash, a letter read, Kannada rendering		about $0.010
Per conversation		**$0.08 to $0.12, about ₹9.5**
Per live minute		**about ₹1.9**

₹9.5
one 5-minute conversation

₹1.9
one live minute

₹209
one daily user per month (22 days)

₹1–3k
a week today, test batteries only

2.3 · Today's spend

The logs show 130 live conversations since 21 Oct, nearly all test batteries, plus the typed banks on the text models at a fraction of a rupee each. That is roughly ₹1,000 to ₹3,000 a week, which is why the prepaid balance ran out once on 17 Oct and has been fine since.

2.4 · At scale, daily users

Assumption: each user has one 5-minute conversation a day on 22 days a month, so ₹209 per user per month. The text-model price rise in January adds about 3%; the band allows 30% for that, for rupee movement and for longer conversations.

Daily users	Per month	Per year	5 years	10 years
1,000	₹2.1 lakh	₹25 lakh	₹1.25 to 1.6 crore	₹2.5 to 3.3 crore
10,000	₹21 lakh	₹2.5 crore	₹12.5 to 16 crore	₹25 to 33 crore
50,000	₹1.05 crore	₹12.5 crore	₹63 to 82 crore	₹125 to 165 crore

2.5 · At scale, if 50,000 means concurrent

> Fifty thousand people speaking to Mitra in the same minute for eight office hours is 24 million live minutes a day. At ₹1.9 a minute that is about **₹1 crore a day, ₹22 to 25 crore a month, ₹270 to 300 crore a year**, and ₹1,400 to 3,000 crore over five to ten years. Nobody should sign that; it is the case for owning.

2.6 · The owning comparison, from the sovereign plan

Scale	Peak live sessions	Sovereign hardware, one time	Gemini, per year	Payback
The pilot	tens	the two H200s we already own	a few lakh	zero API cost from the day the English core is live
10,000 daily users	a few hundred	two to three nodes, ₹3 to 4 crore	₹2.5 crore	about 18 months
50,000 daily users	a few thousand	about 12 nodes, ₹15 crore	₹12.5 crore	about 14 months

After payback the sovereign stack costs power and people only. The audio, the text and the weights never leave the government's own hardware, which no Gemini plan, including Vertex Mumbai with zero retention, can offer.

Sources

Gemini API pricing · PersonaPlex card · Moshi repository · moshi-finetune · Kyutai TTS 1.6B · Kyutai STT 1B · Mimi · VoiceChat 11B card · gpt-oss-120b · Qwen3-30B-A3B · Sarvam-105B FP8 · Open Yap 1K · Smart Turn · Silero VAD · DeepFilterNet · openWakeWord · Piper · Qwen3-ForcedAligner · pyannote community-1 · Parakeet and Canary · Whisper large-v3-turbo · Chatterbox · VibeVoice · Sesame CSM · Orpheus · Granite Guardian · Full-Duplex-Bench · DailyTalk · AMI corpus · MUSAN · RIRS_NOISES · Live API token billing thread · Rupee rate